8 An isotope of boron is used in the treatment of cancer in the brain.

Boron sticks to cancer cells in the brain.

(a) The isotope of boron is bombarded with neutrons then undergoes fission to form lithium and alpha-particles.

(i) Describe **one** difference between fission and fusion.

.....

..... [1]

(ii) A nucleus of boron (B) contains 5 protons and 5 neutrons. Complete the nuclide equation for this fission reaction.

$$\text{.....}^{\text{.....}}\text{B} + {}^{1}_{0}\text{n} \rightarrow \text{.....}^{\text{.....}}\text{Li} + \text{.....}^{\text{.....}}\alpha$$

[3]

(b) The alpha-particles destroy the cancer cells. Suggest and explain **one** reason why alpha particles are more suitable than gamma radiation for use in this treatment of brain cancer.

.....

.....

..... [2]

(c) Other cancers are treated with gamma radiation. Describe **one** safety precaution a nurse or radiologist takes during this treatment.

.....

..... [1]

[Total: 7]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN